Deutsche Bank
Research Institute

Chapter 2: Critical minerals

3 March 2026

- Rare earths are vital components for a state's defense and technology complex
- Without rare earths – and critical minerals more broadly – the three other chokepoints of chips, cables, and cloud cannot function.
- The key bottlenecks are in processing (separation, refining, alloying, magnets), are predominantly located in China.
- Using the historical analogy of Fordism in the 20th century, we look at how critical minerals have evolved from industrial outputs into strategic "chokepoints" across key technology and defense needs.

Our new series, "The Great Rebalancing," focuses on five critical "chokepoints" that are redefining global power: Currencies, Critical Minerals, Chips, Cables, and Cloud.

Each chapter of our series will explore how and why these chokepoints define a new matrix of global power. Currencies allocate risk, critical minerals anchor the physical economy, chips generate compute power, cables move data and finance, and cloud platforms organize markets. Influence over this layered system shapes geopolitical standing – determining which nations can scale their economies, exert control, and apply strategic pressure.

This series argues that a "great rebalancing" is underway, shifting the global order from liberal institutionalism, where interdependence fosters cooperation, toward what we call "infrastructure realism". In this new paradigm, control over key systems and inputs shapes outcomes. Under "infrastructure realism", material and digital infrastructures become decisive instruments of state power. The US-China trade tensions of 2025 starkly illustrated this shift. The financial system — from bond markets to payment rails to digital money —acts as the meta-infrastructure that coordinates and prices all others.

Thus, this series addresses the wider question: which major powers are best positioned for influence and what are their strategic advantages? The 2025 US National Intelligence Annual Threat Assessment identified China as the primary actor challenging US global interests, citing its use of "asymmetric and conventional hard power tactics" and its drive to establish "alternative systems" to compete with the US. While the US-China power competition is longstanding, 2025 highlighted the asymmetric economic dependencies and capabilities shaping the relationship between the two powers.

As these five chapters will demonstrate, power now rests on the control of critical infrastructure. Both the US and China are in a race to reduce strategic dependencies on each other, and the outcome will define the next era.

Authors

Marion Laboure
Senior Strategist

Camilla Siazon
Research Analyst

Deutsche Bank
Research Institute

Introduction: From inputs to chokepoints, the competition for industrial control

In this chapter, we examine how the breadth and scale of the 2025 US-China critical minerals crisis supports our central thesis that the global economic order is moving towards infrastructure-realism. [1] In Chapter 1 of this series, we analyzed this through the lens of currency and payments architecture (see here). In Chapter 2, we demonstrate how control over the industrial processes for rare earths (and critical minerals more broadly) is the next layer in this system of power. This control spans every stage, from mining and processing to setting technical standards and securing offtake agreements. As the world undergoes a rebalancing towards an infrastructure-determined order, it is likely that conflict will accelerate for control over these strategic chokepoints.

Rare earths, often described as the "vitamins" of high-tech manufacturing, are integral components vital for defense systems, electric vehicles, power grids and data centres. Concerns that China could use its rare earth dominance as political leverage are not new; in 2010, for example, Japan faced shortages and price spikes after a halt in shipments from China led many to link the event to a preceding diplomatic clash. [2] In addition, critical mineral self-sufficiency has been many countries' longstanding priority for decades (in particular Europe's), especially as part of their goal towards a net-zero energy transition.

Still, 2025 revealed the extent to which rare earths have become an explicitly binding constraint in US-China relations. Why? The answer brings us to our main argument: rare earths and critical minerals have grown to become indispensable to a country's national security; a hallmark of an infrastructure-realist world.[3] In October 2025, US Treasury Secretary Scott Bessent warned that Beijing's controls pointed "a bazooka at the supply chains and the industrial base of the entire free world." [4]

The financial implications of this are significant. The Atlantic Council, for instance, estimates an effective Chinese export ban on rare earths would lead to severe shortages across the energy, automotive and defense sectors within weeks. It also calculates that a one-year disruption of dysprosium, neodymium, and manganese would reduce US GDP by $1.6bn, $154mn, and $96mn, respectively. [5] Addressing this vulnerability, the 2025 US National Security Strategy identifies critical minerals as a national security chokepoint, arguing that the US "must never be dependent on any outside power" for inputs essential to defense and economic resilience. [6]

The core of the critical minerals crisis is not geological scarcity, but industrial concentration. China's capacity to exert geopolitical influence through its critical minerals sector in 2025 stems directly from its decades-long effort to industrialize its critical minerals supply chains. Today, only China possesses the capacity to process rare earths into high-purity oxides, metals, and magnets at scale. The dependency is stark: the US and the EU import a respective 71% and 46% of their rare earths from China, and the US

---

[1] n our series introduction, we define "infrastructure realism" as a new world order where state control over key systems and inputs shape outcomes. Under infrastructure realism, material and digital infrastructures constitute the decisive instruments of state power.

[2] Previously published: Marion Laboure and Cassidy Ainsworth-Grace, "US vs China: Understanding the Rare Earth Metals Tensions in 10 Charts." Deutsche Bank Research, January 18, 2023, https://research.db.com/research/Article?rid=9d202c96-b32e-4e3d-a0ff-9a418104a6f5 604&kid=RP0001&documentType=R&wt_cc1=IND-2758331-5233; and Marion Laboure and Cassidy Ainsworth-Grace, "Tangled Wires: The Geopolitical Landscape of Semiconductors & Rare Earth Metals," Deutsche Bank Research, June 7, 2023, https://research.db.com/research/Article?rid=e5a16334-9484-490f-9217-7ab2649de709-604&kid=RP0001&documentType=R&wt_cc1=IND-2758331-5233.

[3] A tenuous truce was reached in November 2025, after both sides agreed to a one-year delay for Beijing's rare earth export controls.

[4] Doug Palmer, "Bessent Says US in Talks with China to Prevent New Trade War," Politico, October 13, 2025. https://www.politico.com/news/2025/10/13/bessent-china-talks-trade-00606345.

[5] Reed Blakemore, Alexis Harmon, and Peter Engelke, "Critical Minerals in Crisis: Stress Testing US Supply Chains against Shocks" (issue brief, Atlantic Council, October 9, 2025), https://www.atlanticcouncil.org/in-depth-research-reports/issue-brief/critical-minerals-in-crisis-stress-testing-us-supply-chains-against-shocks/.

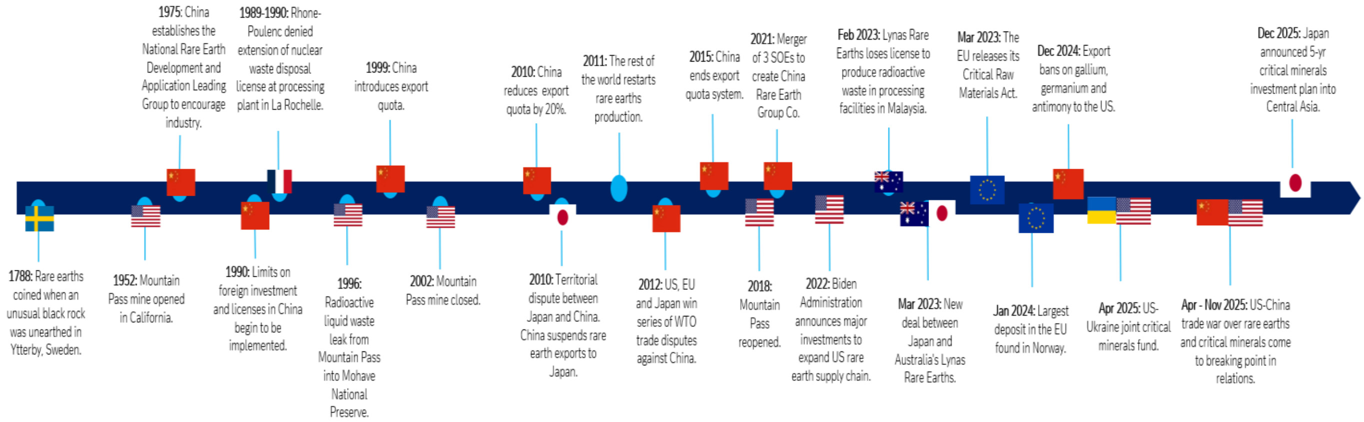
[6] Now under the current administration, the US views mineral self-sufficiency as integral to its current goal of strengthening the country's military industrial complex. This is because the USGS recognizes the US is at least 50% import reliant on many critical minerals essential for defense, including cobalt, graphite, gallium, and several rare earths.

imports 100% of its heavy rare earths from China. [7] Thus, the scale and impact of Beijing's dominance over critical minerals is a clear assertion of infrastructure realism. By establishing and scaling its control over this foundational layer of the physical economy, China has gained a powerful tool to shape outcomes and advance its economic priorities. China's latest strategic move amidst tensions over Taiwan exemplify this, with Beijing restricting "dual-use" rare earth magnets and critical minerals to 20 Japanese companies in February. [8]

This ascension of power via industrialization calls to mind a useful historical analogy: just as US industrial strategy during the period of Fordism in the 20th century drove its leadership in sectors like automobiles and defense, China has strategically focused on industrializing its critical minerals supply chains in the 21st century.

While we anticipate competition in this sphere will become more crowded over the next decade, our analysis suggests that China is well-positioned to remain the dominant actor for the foreseeable future. In response, we expect the US to employ increasingly assertive and protectionist policies as it seeks to reindustrialize its own mineral supply chains. Under this tense push-and-pull, minerals used as leverage will be the foremost feature of the global critical minerals system in the next decade.

Figure 1: Timeline of the global critical minerals industry



Source: Deutsche Bank Research.

1. Critical minerals and rare earths: Laying out the industrial question

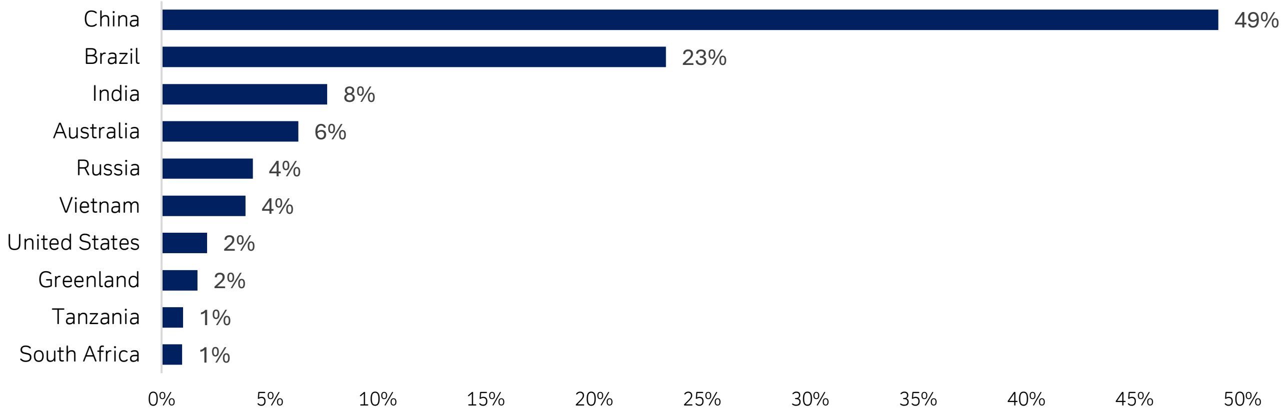
In 1992, Deng Xiaoping famously declared, "the Middle East has oil, China has rare earths." However, unlike oil, where geopolitics are largely shaped by supply volumes and demand, the landscape of rare earths and the broader critical minerals sector is far more complex.

For one, while rare earths and most other critical minerals are not geologically scarce, their upstream supply is highly concentrated. Approximately five countries hold nearly 90% of known rare earth reserves, creating significant vulnerabilities in the global supply chain.

[7] Heavy rare earth metals are a subgroup of rare earths, characterized by higher atomic weights, smaller ionic radii, and greater scarcity compared to light rare earths. They are essential for high temperature magnets in electric vehicles, defense systems, and green technology. "MINERAL COMMODITY." 2026. https://pubs.usgs.gov/periodicals/mcs2026/mcs2026.pdf. For Europe, Eurostat. 2025. "Imports of Rare Earth Elements Saw 30% Drop in 2024." @EU_Eurostat. Eurostat. April 9, 2025. https://ec.europa.eu/eurostat/en/web/products-eurostat-news/w/ddn-20250409-1.

[8] McMorrow, Ryan, Harry Dempsey, Joe Leahy, and Leo Lewis. 2026. "China Slams Dozens of Japanese Companies with Export Curbs." @FinancialTimes. Financial Times. February 24, 2026. https://www.ft.com/content/afa45318-1c56-435a-a3c8-8b82a93714a0?shareType=nongift.

Figure 2: Mineable rare earths reserve as a % of total global minable reserves



Source: Deutsche Bank, USGS.

Even more pronounced is the concentration of midstream processing capacity. This is because the core challenge with critical minerals lies not in their extraction, but in their transformation into usable, high-value industrial inputs. The capital, infrastructure, and skilled labour necessary to extract, separate, and refine critical minerals are considerable and slow to scale. As we will demonstrate in the next section, it is Beijing's dominant position in these industrial processing stages that has enabled it to convert its technical dominance into geopolitical leverage.

1.1 The material base of modern power: Why critical minerals matter…

A mineral or metal is deemed "critical" if it meets two criteria: (i) it is of high economic or strategic importance, particularly for the energy transition, defense, technology, and industrial production; and (ii) its supply chain is vulnerable to disruption, either due to foreign dependence, geographic concentration, limited substitutability, or geopolitical risks. Criticality, therefore, is a measure of strategic vulnerability, not of geological scarcity.

Over time, the list of materials falling under this definition has expanded to include lithium, cobalt, nickel, copper, graphite, gallium, germanium, and tungsten. While many of these metals have been in use for centuries, their strategic importance grew significantly during the 20th century with the rise of industrial warfare and mass electrification. In the 21st century, critical minerals have become foundational to national security, evolving from mere inputs for consumer goods to core components of military, technological and industrial infrastructure.[9] Below are some examples of how critical minerals have come to represent a strategic chokepoint, and how they are essential for the three other layers we discuss in this series (chips, cables and cloud). [10]

- Defense: According to defense analytics firm Govini, 80,006 components across 1,908 U.S. weapons platforms rely on minerals subject to Chinese export controls.[11] Breaking this down across weapon systems, this dependency affects 92% of US Navy systems, 85% of the Air Force, 70% of the Army, and 62% of the Marine Corps.
- Advanced technology: Critical minerals enable digital performance, energy efficiency, data storage, and high-speed connectivity. In semiconductors, these

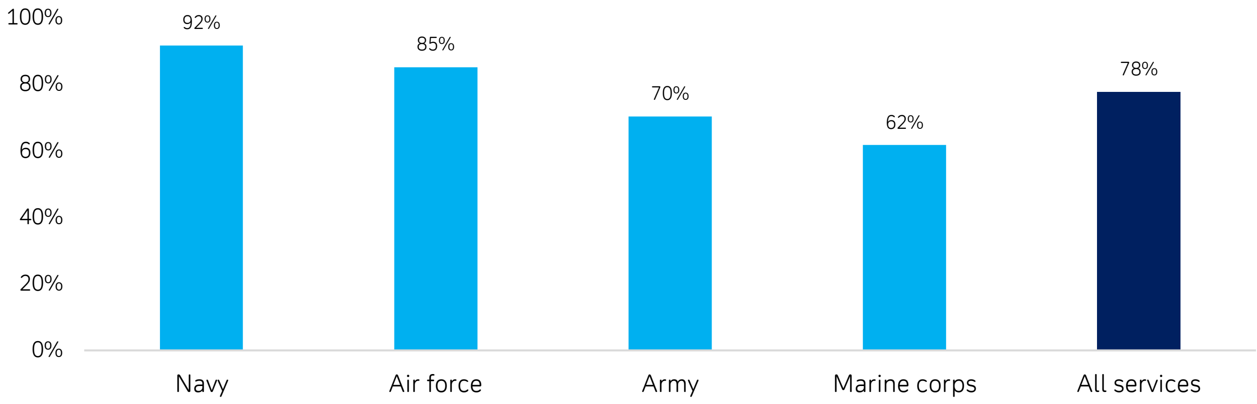
[9] While early lists were concise, the European Union's list of critical raw materials, first published in 2011 with 14 materials, expanded to 34 by 2023. Similarly, the United States' formalization of its critical minerals list has seen significant changes even in recent years. The 2025 list of critical minerals, has now risen to 60. Included the addition of 10 mineral commodities that were not in the 2022 edition—boron, copper, lead, metallurgical coal, phosphate rock, potash, rhenium, silicon, silver, and uranium.
[10] Advanced semiconductors rely on gallium, germanium, tungsten, and rare earths. Cables: Rare earth magnets, specialty alloys, and energetic materials underpin missiles, radars, submarines, and aircraft. Cloud: Data centres' energy and storage infrastructure depend on copper, rare earths, and battery minerals.
[11] Govini, "From Rock to Rocket: Critical Minerals and the Trade War for National Security," 2025, https://www.govini.com/insights/from-rock-to-rocket-critical-minerals-and-the-trade-war-for-national-security.

metals are essential for processing, storing, and transmitting information. For example, gallium – for which China accounts for >99% of global supply – is a crucial material for powering chips in smartphones, satellite systems, 5G infrastructure, power electronics, and AI hardware.[12]

- Clean tech and net zero transition: The transition to electric vehicles, renewable energy generation and electrified infrastructure is heavily dependent on a stable supply of critical minerals. Minerals such as lithium, nickel and cobalt are crucial to battery performance, while rare earth magnets are indispensable for wind turbines and EV motors.

Ultimately, critical minerals are a foundational layer of modern economic, technological, and military power. Demand for these materials tends to rise in tandem with geopolitical tension, as access becomes constrained. This dynamic is central to the current era, as nations like the US seek to reinforce their industrial and defense capabilities.

Figure 3: Critical minerals in US weapons systems, 2025



Source: Deutsche Bank Research, Govini.

1.2 ... and especially rare earth metals

Rare earth metals, a subset of critical minerals, is a group of 17 metallic elements that possess magnetic and conductive properties that have significantly improved the performance, efficiency, longevity, and reliability of countless modern technologies.

First discovered in Sweden in 1788, it was not until the 20th century that their commercial applications became ubiquitous, used across a wide variety of intermediate and consumer products. Today, the US Geological Survey (USGS) identifies rare earths as essential components in over 200 commercial products, ranging from consumer goods like mobile phones, computer hard drives, cameras, floodlights, and microwave filters, to industrial applications such as petroleum refining.[13]

The primary strategic chokepoint in the rare earth supply chain is in processing. The unique chemical characteristics of rare earths make the separation stage of the supply chain a technologically demanding and capital-intensive endeavour requiring highly specialized expertise. As Thyssen and Binnemans (2011) show in their scientific study on rare earths, these minerals have extraordinarily similar chemical and physical properties, leading to a decades-long struggle for scientists to isolate and place them on the periodic table. This similarity renders traditional chemical extraction methods ineffective.

Producing high-purity rare earth oxides – and subsequently, metals and alloys – requires complex extraction and waste handling, making the process capital-intensive,

[12] "Critical Minerals in Artificial Intelligence," SFA (Oxford), 2025, https://www.sfa-oxford.com/knowledge-and-insights/critical-minerals-in-low-carbon-and-future-technologies/critical-minerals-in-artificial-intelligence/.
[13] Marion Laboure and Cassidy Ainsworth-Grace, "US vs China: Understanding the Rare Earth Metals Tensions in 10 Charts," Deutsche Bank Research, January 18, 2023.

environmentally sensitive and technologically demanding.[14] As a result, many countries can mine rare-earth metals, but China is currently the only nation with the capacity to separate and process them at a mass industrial scale.

2. China's industrialization of critical mineral supply chains: A historical perspective

No country has capitalized on the critical minerals era more effectively than China. Through a long-term, state-directed industrial strategy and significant investment, China now exercises a near-monopoly over key mineral supply chains, turning industrial scale into a source of significant leverage and a bottleneck for the rest of the world. This bottleneck is particularly challenging to overcome because it is rooted in technological capacity, is capital-intensive, and is slow to replicate. Focusing on rare earths as the primary example, the remainder of this section explains how China built, consolidated, and ultimately learned to leverage its industrial advantage in the geopolitical sphere.

2.1 China's midstream dominance: the key is capacity, more than reserves

Up until the late 1980s, the United States was the global leader in rare earth production. The discovery of a large, high-grade rare earth deposit at Mountain Pass, California in 1949 paved the way for mass production of commercial quantities of rare earth elements, which in turn reduced prices and spurred their wider commercial application.[15] France was also once at the forefront of rare-earth processing via its chemical giant Rhone-Poulenc (later acquired by Solvay).

However, western advantage was overtaken as China's leadership recognized the strategic value of its own vast rare earth reserves and launched a concerted effort to develop its domestic production capacity. To do this, the state supported programs that would work on enhancing mining techniques, invested in R&D for rare earth applications, and provided financial backing from state-controlled banks, allowing domestic rare earth companies to operate and scale even without immediate profitability. As a result, the country's domestic production grew at a rapid average of 40% annually from 1978 to 1989.[16]

As China's rare earth exports increased over the 1990s, and the United States began to prioritize environmental and labor concerns that made rare earth processing less viable, production steadily shifted to China as it became increasingly difficult for non-Chinese producers to compete on price. Globalization, combined with China's emergence as the world's workshop, further encouraged the US and Europe to step back from domestic production as supply chains migrated to China. With cheaper domestic supplies at their disposal, Chinese manufacturers also had the opportunity to scale the production of key end-products such as high-strength magnets and electronics, crowding out foreign competition in downstream manufacturing industries as well.

To consolidate its advantage in rare earth production, authorities simultaneously reduced their export quotas – by 54% between 2005-2010 – to secure resources for domestic demand and regain control over operations.[17] This caused international prices for rare earths to increase dramatically, only for Beijing to later release significant volumes of subsidized output to the markets. This self-reinforcing cycle created an oversupply of unprocessed materials, drove down prices, and solidified China's position as the dominant buyer and processor.

---

[14] Pieter Thyssen and Koen Binnemans, "Accommodation of the Rare Earths in the Periodic Table: A Historical Analysis," in *Handbook on the Physics and Chemistry of Rare Earths*, vol. 41 (Elsevier, 2011) 1-93, Why Rare Earths Aren't Really "Rare"-and Why China Still Controls the Choke Point and Accommodation of the Rare Earths in the Periodic Table: A Historical Analysis. As Thyssen and Binnemans also document, many early rare earths OR elements were in fact complex mixtures: metals such as "ceria" and "yttria" took over a century to fully deconstruct, while "didymium" was later separated into Praseodymium and Neodymium.

[15] "Rare-Earth Metal Prices in the USA Ca. 1960 to 1994," USGS, 2026, https://www.usgs.gov/publications/rare-earth-metal-prices-usa-ca-1960-1994-0.

[16] "China's Rare Earths Industry and Its Role in the International Market," U.S.-China Economic and Security Review Commission, November 2010, https://www.uscc.gov/sites/default/files/Research/RareEarthsBackgrounderFINAL.pdf.

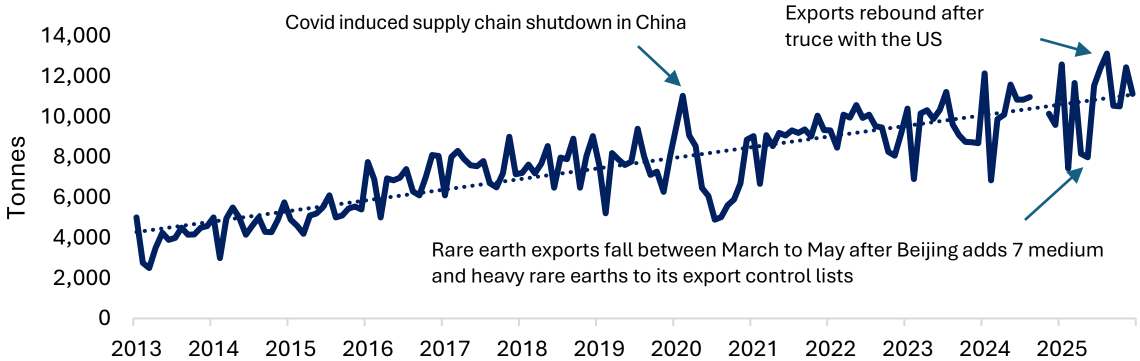
[17] Ibid.

China's control over global prices is reinforced by a lack of external visibility into true rare earth and critical mineral pricing. This is because China's two domestic rare earth exchanges operate with limited transparency, while the Shanghai Metals Market publishes indicative price ranges rather than real-time spot prices. Outside of China, the OTC market for these metals is highly opaque, and trading volumes on the London Metal Exchange are virtually non-existent. The cessation of public reporting on China's rare earth mining and refining quotas further contributed to price uncertainty and corporate volatility as foreign companies struggled to navigate the less transparent environment.

As a result of these factors, China has dominated rare earth processing conversion capacity since the 1990s, with its share of global production peaking at 97.5% in 2009. Today, China maintains dominant positions in rare earth mining (~60–70%), rare earth processing (~90%), and magnet manufacturing (~93%). However, while the country overwhelmingly dominates the midstream and downstream segments, it still relies on large volumes of raw metal imports to refine and process. Thus, China is also one of the largest consumers of raw element imports across critical mineral supply chains. To close this gap, Beijing has expanded its strategic stockpiles and increased overseas investments in mines (e.g. Indonesia).[18] The record $213.5bn investment in the Belt and Road initiative in 2025—including $32.6bn directed to metals and mining to secure processing capacity and offtake rights – is a recent example of this strategic focus. Through these initiatives, China is also strengthening its position in the upstream segment, moving towards even greater influence across the entire critical mineral value chain.

Finally, over the years, Beijing has consolidated state-owned enterprises in the rare earths industry, enabling vertical integration to control the full value chain. In December 2021, Beijing announced the creation of China Rare Earth GroupCo, merging three major state-owned enterprises and two research institutions. This new entity now controls roughly 70% of China's domestic rare earth production.

Figure 4: China rare earth exports



Source: Deutsche Bank Research, China General customs. Note: Data for Sep and Oct 2024 not available.

2.2 Critical minerals as "Mineral Fordism"

To understand the relationship between China's dominance in critical minerals infrastructure and its role in the international order, it is useful to draw an analogy with the global era of American industrialization in the 20th century, a system often referred to as "Fordism".

Fordism was a system of mass production and consumption in the 1920s that relied on standardized parts and moving assembly lines. Its defining feature was vertically integrated production at scale (e.g., the assembly line). Using Stefan Link's definition in *Forging Global Fordism*,[19] Fordism also functioned as a tool of state-led modernization,

[18] Marion Laboure and Camilla Siazon, "The Top 10 Strategic Commodities to Watch," Deutsche Bank Research Institute, October 22, 2025. https://www.dbresearch.com/PROD/RIPROD/PDFVIEWER.calias?pdfViewerPdfUrl=PROD0000000000606588.
[19] Stefan J. Link, "Forging Global Fordism: Nazi Germany, Soviet Russia, and the Contest over the Industrial Order," (Princeton: Princeton University Press, 2020).

enabling countries to accelerate their industrial development and rearmament in the lead-up to World War II.

This dynamic was evident in how both Germany and the Soviet Union actively sought American technology and production methods to rapidly modernize their economies and spread their own state-led ideologies. As Link demonstrates, the open patent system and permissive technology transfer between the US and the USSR and Germany during this time allowed engineers from these countries to study, copy and replicate the methods from the Fordist production systems in Detroit to ultimately build their own industrial and military capabilities.[20] Fordism's mass industrialization, combined with the US's post-war soft power, consumer capitalism and embedded liberal institutionalism, helped cement US hegemony in the 20th century.[21]

Just as Fordism underpinned the US's economic and military rise, China has managed to pioneer and master a new production system for critical minerals that now forms the basis of its geopolitical influence. Through the vertical integration of rare earth companies and dominance as the world's largest downstream off-taker, China has built the world's most complete processing ecosystem for critical minerals thus far. And just as Ford standardized physical parts by achieving massive economies of scale and driving down costs, China has effectively set global prices and production standards. This is most evident in its pricing power over many critical minerals, which has made it the most rational choice for Western companies to mine raw materials globally and ship them to China for processing.

Today, the US is seeking to re-industrialize its own rare earth supply chains but faces what Link describes as the constraints of the "late developer." Unlike the tech transfers that took place between the US and Germany and the Soviet Union during Fordism, in today's infrastructure-realist world, there appears to be less scope for such cooperation between strategic competitors. The West cannot simply replicate China's industrial model without also absorbing the immense economic, social, and environmental costs that China accepted during its own decades-long supply-chain transformation.

2.3 China's critical minerals leverage as a tool of infrastructure realism: a rare earths case study

China's rare earths strategy represents a modern form of "mineral Fordism." In this section, we examine how its near-total dominance over rare earths and critical minerals has enabled the country to leverage this industrial advantage for geopolitical purposes.

While China has long recognized the advantage of its rare earth dominance and has periodically used export quotas and restrictions since the 1990s, we argue that its initial aim for these controls was to strengthen its own domestic capabilities rather than to exert any geopolitical influence.

The example many cite as the first time China used its market dominance for political leverage was in September 2010, when shipments of rare earths from China to Japan were halted for two months. Many observers concluded that this was a retaliatory measure, since the disruption occurred soon after a hostile clash between Chinese and Japanese boats in the long-disputed waters near the Senkaku Islands (called the Diaoyu Islands by China). Contrary to popular narrative, however, there is little evidence that China publicly announced a targeted export ban on Japan in 2010. While China threatened "strong countermeasures" against Japan, it never formally confirmed exports cuts of rare earths to Japan,[22] leaving analysts to infer intent from trade data. China's premier at the time also

---

[20] Ibid. According to Link, Germany ultimately internalized these methods more effectively than the Soviet Union.
[21] Ibid.
[22] A study by Simon Evenett and Johannes Fritz found that while China certainly applied export restrictions on rare earths during the 2010s, there was no evidence of a targeted ban on Japan. In fact, it was Australia, not Japan, that witnessed sharp falls in monthly REM exports from China during this period. Simon Evenett and Johannes Fritz, "Revisiting the China–Japan Rare Earths Dispute of 2010," *CEPR*, July 19, 2023. https://cepr.org/voxeu/columns/revisiting-china-japan-rare-earths-dispute-2010.

said that “China is not using rare earths as a bargaining chip.”[23] Nevertheless, the market impact was clear: rare earth metal prices surged nearly tenfold following the disruption. Afterwards, Japan (in conjunction with the US and EU) filed a successful WTO trade complaint alleging China’s use of export quotas and other restrictions on rare earths, leading China to dissolve any restrictions by 2015.

The more systematic use of rare earths as leverage began as US-China competition intensified during the first Trump administration. Then, Beijing’s export restrictions on its rare earths and critical minerals became more systematic and concentrated. Indications that the government was beginning to view its rare earth dominance as a strategic tool for national security emerged in 2019, when tensions over the US-China trade war reached a new peak. During a high-profile visit to one of China’s major rare earths mining and processing facilities in Ganzhou, President Xi Jinping made remarks to state media that rare earths could be a “new Long March” battleground.[24] This episode can be interpreted as the first informal signal that Beijing had started to consider its near-monopoly over the industry as a point of leverage in its trade dispute with the US. These concerns were amplified one year later when China’s National People’s Congress passed a new export control law, creating a legal framework to restrict exports of sensitive materials on national security grounds. While rare earths were not named outright, the new law allowed Beijing to legally halt its exports to specific countries if it chose to do so.

By the 2020s, as global competition evolved into a tech race, China’s actions became more explicit. In 2023, it banned the export of specialized rare earth processing equipment and imposed export restrictions on gallium, germanium and certain types of graphite.[25] In 2024, these restrictions were tightened into a full export ban of gallium and germanium to the US. By 2025, China had developed a range of policy tools related to critical minerals, effectively transforming its industrial dominance into a bargaining tool.

[23] Su Qiang, “Rare Earth Will Not Be Used as Bargaining Chip,” *China Daily*, October 8, 2010.
[24] The “Long March” comment is seen as a reference to the massive, year-long tactical retreat by the CCP in 1934, as the Red Army escaped the KMT forces during the Civil War. It cemented Mao Zedong’s leadership, allowed the weakened party to survive, and is now celebrated as a foundational symbol of perseverance, sacrifice, and strength. Xi’s visit occurred soon after the US had backlisted Huawei and came amidst fears that Beijing would restrict or embargo rare earth exports to the US. Zhou Xin, Wendy Wu, and Kinling Lo, “Xi Jinping Visits Rare-Earth Minerals Facility amid Talk of Use in US Trade War,” *South China Morning Post*, May 20, 2019. https://www.scmp.com/economy/china-economy/article/3010977/xi-jinping-visits-rare-earth-minerals-facility-amid-talk-use.
[25] Marion Laboure and Cassidy Ainsworth-Grace, “Rare Earth Metals: What to Expect in 2024 in 10 Slides,” Deutsche Bank Research, March 26, 2024, https://research.db.com/research/Article?rid=eec2bc46-c2ac-4f07-bcbe-2c75d836a3ee-604&kid=RP0001&documentType=R&wt_cc1=IND-3146993-0000.

Figure 5: How China controls the whole value chain

Mineral	Use cases	Upstream dominance (mining & reserves)	Midstream dominance (refining & processing)	Key dynamics & strategic actions
Nickel	EV batteries, defence alloys, aerospace materials.	Indonesia holds the largest reserves and leads mined production (67%), but China controls the value chain.	China controlled 65% of global refined nickel output in 2024, primarily through ownership of Indonesian refining capacity.	China, the largest nickel consumer, is increasing stockpiles due to escalating US trade tensions.
Lithium	Lithium-ion batteries for smartphones and laptops, power tools and grid-scale energy storage systems.	Australia, Chile, and China produced ~72% in 2025. China controls 21% of upstream supply.	China controls ~70% of lithium chemical refining. The top three refiners (China, Argentina, Chile) account for ~95% of global capacity.	China's control over the lithium value chain is growing. The US imported over 90% of its lithium-ion battery cells from China in 2024. US response: investment in Canadian mining (e.g., Thacker Pass Mine). In the UK, the country's first commercial lithium plant opened in Feb 2026.
Cobalt	EV and defence batteries, jet engine alloys, industrial gas turbines, electroic warfare systems.	~74% of mined cobalt comes from the DRC, but the IEA projects Indonesia to bring its market position down to 50% by 2040. Chinese companies owned 80% of DRC production in 2024 (EIA).	China refined ~78% of global cobalt market in 2024; primarily for lithim-ion batteries.	China remains the top buyer, driven by its battery industry. The US was 79% net import-reliant for cobalt in 2025. US response: the US is diversifying supply chains with new mining and processing projects, facing challenges from weak prices (e.g., US stake in Trilogy Metals in Alaska).
Copper	Aircraft, ground vehicles, ships, submarines, missiles, AI data centers.	Chile has the world's largest reserves and leads global mining production (23%).	China dominates downstream supply chains, controlling 48% of global refining capacity.	US response: the Trump administration has designed a copper policy to onshore mining and refining production in the US.
Gallium & Germanium	Essential semiconductor materials used in fiber optics, infrared optics, solar. technologies, advanced military and biomedical applications.	China controls 99% of gallium production and nearly all of germanium production.	Control over high-purity refined output used for chips and defence is more balanced, with Europe, the US, Japan and the Canada overseeing some parts of the midstream segment.	No US domestic gallium production has existed since 1987, while some limited germanium production occurs. The US also stockpiles germanium in its National Defense stockpile, but no gallium. In 2025, the DoE announced ~$6mn in funding for gallium.
Tungsten	Steels, aerospace, and military projectiles.	China controls 82% of Tungsten production; also has the world's largest Tungsten reserves.	China refines over 80% of the world's tungsten concentrate into intermediate products.	Tungsten concentrate production ex- China = 21% of total world production. China remains the top consumer of Tungsten. In 2025, China placed several restrictions on critical minerals including Tungsten to the US.
Rare earths	17 minerals critical for defence and tech – F-35 fighter jets, submarines, Tomahawk missiles, radar systems, unmanned aerial vehicles etc), and advanced semiconductor technology.	China dominates the REM supply chain, controlling ~69% of rare earth mining in 2025.	China controls >90% of refining capacity.	IEA projects China will still dominate refining (76%) and mining (51%) by 2030, implying rare earths should remain the frontline in the US-China resource war.
Graphite	Batteries, brake linings, lubricants, powdered metals, refractory applications, and steelmaking.	China controls 78% of graphite production. The US has had no domestic natural graphite production since the 1950s.	China maintains dominance across mining, processing, and anode manufacturing.	The US and EU import around 46% (2021-2025) and 38% (2024) of their graphite from China, respectively. China requires export permits for high-grade graphite (Dec 2023), while the US imposed 93.5% anti-dumping duties on Chinese graphite and anode materials (July 2025). US response: the US is developing its upstream supply chains; the USGS currently tracks five active exploration or development projects in the US.

Sources: Deutsche Bank Research, USGS, IEA, EIA.

3. The new mineral map: Bloc formation and infrastructure partnerships

In response to supply disruptions and heightened geopolitical tensions surrounding critical minerals in 2025, governments and companies have stepped up their efforts to restructure their critical minerals supply chains. What is emerging is not diversification in the classical sense, but a re-mapping of supply chains along strategic, security-aligned blocs. Thus far, the following themes have emerged, characterized by uneven progress and significant constraints.

3.1 State consolidation

Similar to China's approach to vertical integration, Washington is seeking partial state ownership in domestic critical mineral companies. Since January 2025, the Trump administration has announced multiple partnerships with domestic and international minerals companies as it seeks to build private sector willingness to invest. Recent examples include a 10% stake in Trilogy Metals in Alaska to target cobalt, copper and related minerals; a $150mn stake in a Louisiana-based gallium producer;[26] a 5% stake in the Canadian company Lithium Americas; and a 5% stake in its Thacker Pass mine in Nevada, scheduled to open in 2028. Notably, in July 2025, the Pentagon also became the

[26] Steff Chávez and Camilla Hodgson, "Pentagon Invests $150mn in US Gallium Company to Secure Strategic Supplies," *Financial Times*, January 12, 2026, https://www.ft.com/content/00b2ba44-9f20-4d41-9b01-3df9dde5dae7.

largest shareholder of MP Materials. In response to these policy efforts, shares of MP Materials, Lynas, and USA Rare Earths soared in 2025.

The US is not the only country pursuing this strategy; other countries such as Australia and Canada are implementing similar measures to encourage private sector investment. Just as the Chinese state supported its nascent rare earths industry during the 1980s, so too are other governments today. This intervention aims to yield notable dividends by countering the historical private sector reluctance to build or maintain capacity, a hesitation often influenced by significant price fluctuations, at times exacerbated by supply dynamics originating from China, which has often led to facility closures and threatened the viability of Western domestic mining, processing and downstream manufacturing capacity.

3.2 Stockpiling

In response to supply chain risks, the US and the EU, among others, have announced or expanded critical minerals stockpiles. However, this strategy has yielded limited gains. Many speciality critical minerals cannot be effectively stockpiled in their final, usable forms because they are alloyed into highly specific forms and combinations for diverse end-uses. As such, existing public and private stockpiles of key processed materials, like neodymium or dysprosium, would be depleted within weeks or months under an effective export ban of critical minerals.[27] We saw this play out in April 2025, when automakers raised alarms over rare earth shortages following China's implementation of export controls, with Ford's CEO describing US dependency on Chinese rare earths as a "hand-to-mouth situation."[28]

Ultimately, stockpiling raw ore does not solve the core problem. The binding constraint remains midstream processing capacity, not geological resources: without maintaining the full value chain, stockpiling offers limited protection. The Atlantic Council notes that even "sufficient" raw inputs stockpiles may be ineffective if midstream bottlenecks persist. It also argues that stockpiles are more effective at derisking new domestic minerals projects and countering price manipulation than filling market gaps during a major supply crisis.[29]

3.3 Mineral deals as the new security treaties

The third emerging theme we have detected is the evolution of mineral alliances beyond traditional multilateralism, moving towards arrangements with strategic security implications. This strategic reorientation is largely led by the US, as evident in its latest National Security Strategy, which states "America's economic partners should no longer expect to earn income from the United States through overcapacity and structural imbalances but instead pursue growth through managed cooperation tied to strategic alignment and by receiving long-term U.S. investment."[30]

For the US, these allied mineral deals serve multiple objectives:[31] gaining access to partner stockpiles (e.g. in Japan and South Korea), forming buyers' consortia to increase purchasing power, and using trade agreements to insulate domestic firms from price volatility. This dynamic has been visible in recent US engagement with Ukraine and Greenland.

---

[27] Reed Blakemore, Alexis Harmon and, Peter Engelke, "Critical Minerals in Crisis: Stress Testing US Supply Chains against Shocks," (issue brief, Atlantic Council, October 9, 2025), https://www.atlanticcouncil.org/in-depth-research-reports/issue-brief/critical-minerals-in-crisis-stress-testing-us-supply-chains-against-shocks/.
[28] "'Hand-To-Mouth': Ford CEO Confirms Factory Shutdown amid China's Rare Earth Chokehold," *Rare Earth Exchanges*, June 14, 2025, https://rareearthexchanges.com/news/hand-to-mouth-ford-ceo-confirms-factory-shutdown-amid-chinas-rare-earth-chokehold/.
[29] Reed Blakemore, Alexis Harmon and, Peter Engelke, "Critical Minerals in Crisis: Stress Testing US Supply Chains against Shocks," (issue brief, Atlantic Council, October 9, 2025), https://www.atlanticcouncil.org/in-depth-research-reports/issue-brief/critical-minerals-in-crisis-stress-testing-us-supply-chains-against-shocks/.
[30] "National Security Strategy of the United States of America," The White House, 2025, https://www.whitehouse.gov/wp-content/uploads/2025/12/2025-National-Security-Strategy.pdf.
[31] Reed Blakemore, Alexis Harmon and, Peter Engelke, "Critical Minerals in Crisis: Stress Testing US Supply Chains against Shocks," (issue brief, Atlantic Council, October 9, 2025), https://www.atlanticcouncil.org/in-depth-research-reports/issue-brief/critical-minerals-in-crisis-stress-testing-us-supply-chains-against-shocks/.

However, in an environment characterized by increased national resource focus and infrastructure realism, such multilateral partnerships have clear limitations during a crisis. Potential partner nations may be constrained by their own domestic needs and unable to act as a supplier for the US. Also, many potential allies remain indirectly dependent on Chinese flows in one way or another, which could limit their willingness to participate fully.

3.4 Resource nationalism and export controls

Other countries with significant stakes in critical mineral mining are increasingly imposing export restrictions (e.g., licensing requirements, taxes, quotas, and outright bans) on their upstream raw materials, with the aim of protecting and developing their domestic downstream industries.[32] This trend is significant, as it could complicate China's own strategy of securing upstream raw mineral inputs from abroad.

A notable example is Indonesia, which holds the world's largest nickel reserves. It banned the export of raw nickel ore in 2014 to promote domestic industrialization. However, many resource-rich countries are still unable to replicate this strategy as they lack the processing self-sufficiency. Below, we survey other key players in the critical minerals race and assess their progress and constraints thus far.

3.5 United States: Mineral diplomacy meets a renewed Monroe Doctrine[33]

The US has sought to diversify its critical mineral supply chains away from China since the 2000s. For example, the Biden administration took proactive measures such as funding commercial-scale rare-earth oxide production and supporting MP Materials in 2022. However, unlike the Biden administration, which made a clean energy transition a key goal, the second Trump administration views critical mineral sufficiency as integral to its goal of strengthening the US military industrial base. This is again a characteristic of the infrastructure realism paradigm discussed earlier, where the struggle for critical minerals is increasingly perceived as a contest for global hegemony. Examples that demonstrate how critical minerals are becoming increasingly entwined with US defense include the "One Big Beautiful Bill Act" (July 2025), which allocated $9.5bn to domestic critical mineral capabilities, including $7.5bn for Defense and $2bn for energy dominance and Defense Production Act purposes. Later in October 2025, the Pentagon also announced plans to procure up to $1bn in critical minerals.

This convergence of critical mineral sufficiency and US military security perhaps explains why amongst its many domestic supply chain solutions, the current administration has also embarked on assertive geopolitical expansion to strengthen its minerals industrial policy. This is a marked departure from the multilateralism of economic security guarantees that defined the previous order of "Liberal Internationalism," and a clear step toward infrastructure-realism. The first clear example of this was in February 2025, when Washington said it would not continue supporting Ukraine's war efforts against Russia, unless compensated with Ukraine's vast natural resources.[34] This position reportedly triggered strong reactions from Ukraine, with President Zelensky stating that would be akin to "selling a country." The situation escalated dramatically when the US temporarily suspended all military aid and intelligence sharing to Ukraine in March. But by April, both sides had come to an agreement and a joint infrastructure fund was established.[35] Under

---

[32] David Guberman, Samantha Schreiber, and Anna Perry, "Office of Industry and Competitiveness Analysis Export Restrictions on Minerals and Metals: Indonesia's Export Ban of Nickel," (working paper, Office of Industry and Competitiveness Analysis, U.S. International Trade Commission, 2024)
https://www.usitc.gov/publications/332/working_papers/ermm_indonesia_export_ban_of_nickel.pdf

[33] The Monroe Doctrine was a foundational policy during the early 1800s that prioritized US sovereignty over the Western Hemisphere from European influence; in the 21st century.

[34] Seb Starcevic, "Trump Demands $500B in Rare Earths from Ukraine for Continued Support," *POLITICO Pro*, February 2025. https://subscriber.politicopro.com/article/2025/02/trump-demands-500b-in-rare-earths-from-ukraine-for-continued-support-00203529.

[35] Marion Laboure, Peter Sidorov, and Camilla Siazon, "Thematic Research: US-Ukraine Deal: Strategic Win and Security Tradeoff?," Deutsche Bank Research, May 1, 2025, https://research.db.com/research/Article?rid=d4e3ea2a-dd1a-482b-83e3-e627c5b9d033-604&kid=RP0001&documentType=R&wt_cc1=IND-3146993-0000 .

the agreement, Ukraine committed to direct 50% of its royalties, rent payments, and license fees from its critical minerals' assets to the US.

This episode represents a shift in US foreign policy from traditional aid to economic statecraft, a hallmark of infrastructure realism. In 2026, the Trump administration accelerated its pursuit of critical mineral security by renewing its call to purchase Greenland, which holds the world's eighth-largest rare earth deposit and other critical minerals. President Trump has consistently framed Greenland as vital to US national security, first proposing its purchase in 2019. And while the Trump administration has not formally designated Venezuela as a strategic mineral priority after US forces entered the country and deposed its president in January 2026, US officials have signalled interest in Venezuela's mineral endowment, citing Venezuela's steel and critical minerals as potential areas of US-Venezuela collaboration.[36] These incidents underscore a US dual strategy of securing critical minerals, as well as exploring opportunities for expanded influence or access to strategically important regions.

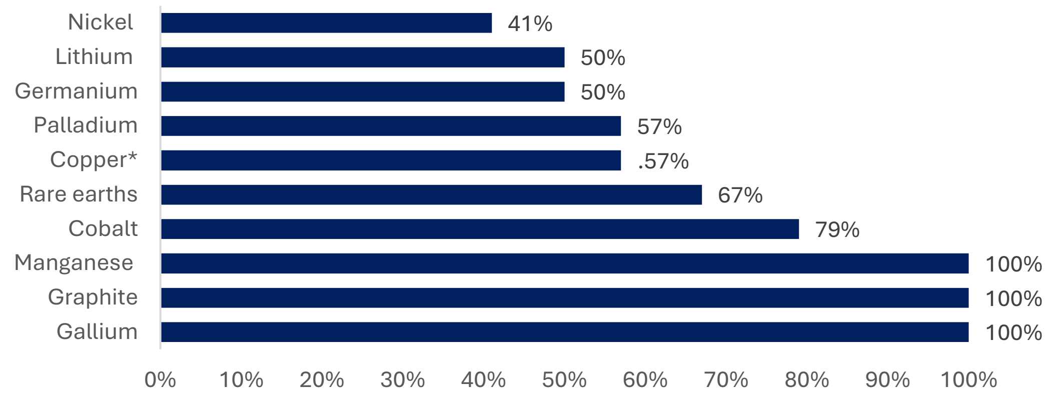
3.6 How effective has the US response effort been?

In October 2025, Secretary Scott Bessent suggested US dependence on China's rare earths could be reduced within 18–24 months.[37] In practice, however, a typical rare earth project requires around 9 years, from exploration to construction.

Thus, US progress is still years away from building sufficient critical mineral chains, with its midstream and downstream capacity continuing to be the primary bottlenecks. Its biggest rare earth company investments, Lynas and MP Materials, still lag on processing capabilities, and according to CSIS, MP Materials' output of NdFeB magnets of 1,000 tons by end-2025 represents less than 1% of China's 2024 output.[38]

In January 2026, the Trump administration pivoted away from tariffs toward international price coordination, directing USTR and Commerce to pursue price floors with partners.[39] This underscores the US's recognition that domestic capabilities alone are insufficient. In his Executive Order, President Trump said that if price floor agreements have not progressed by 13 July 2026, he may implement additional measures including import restrictions and minimum import prices for certain types of critical minerals.

Figure 6: US net import reliance as a % of consumption in 2025, select critical minerals



Source: Deutsche Bank Research, USGS. Note: Copper refined.

---

[36] "Trump wants Venezuela's oil. What about minerals?," Politico Pro, January 2026, https://subscriber.politicopro.com/article/eenews/2026/01/06/trump-wants-venezuelas-oil-what-about-minerals-00712231 . One day after US forces captured Venezuelan President Nicolas Maduro, Commerce Secretary Howard Lutnick said: "you have steel, you have minerals, all the critical minerals" that Trump would "fix and bring it back – for the Venezuelans."

[37] Demetri Sevastopulo, "China 'Made a Real Mistake' by 'Firing Shots' on Rare Earths, Says Scott Bessent," *Financial Times*, October 31, 2025, https://www.ft.com/content/399ad440-ecd3-4fb0-b97d-a0d8e0a8a22c.

[38] Gracelin Baskaran and Meredith Schwartz, "The Consequences of China's New Rare Earths Export Restrictions," CSIS, April 14, 2025, https://www.csis.org/analysis/consequences-chinas-new-rare-earths-export-restrictions.

[39] Ernest Scheyder and Kanishka Singh, "Trump Says No Critical Minerals Tariffs for Now, Will Seek Overseas Supplies," *Reuters*, January 14, 2026, https://www.reuters.com/world/china/trump-says-no-critical-minerals-tariffs-now-will-seek-overseas-supplies-2026-01-14/.

3.7 European Union: The challenge of re-industrialization

Once a leader in the critical minerals supply chain, the EU's position has been significantly affected by the intensifying competition between the US and China, with Industry Commissioner Stephane Sejourne describing the bloc as "collateral damage".[40] According to a recent report from the European Court of Auditors, the bloc is import-reliant on China for several critical minerals, including Gallium (71%), Magnesium (76%) and Germanium (45%).[41]

As the critical minerals crisis intensified in 2025, Europe accelerated its earlier moves for self-sufficiency but has been slower than the US in assembling the necessary toolkit. Unlike the US, the EU continues to place significant emphasis on recycling initiatives and government funding. The most recent measures include the Critical Raw Materials Act (2023) and ReSourceEU (2025), a ~€3bn initiative that builds on the CRMA's goals[42] to target critical raw material extraction (>10%), processing (40%), and recycling (15%) by 2030.[43] Under CRMA, the EU strives to ensure it limits its import reliance from a single third country below 65%.

ReSourceEU will help with its focus on preventing scrap aluminium from leaving the bloc and recycling magnets used in car batteries, supported by a €2bn annual fund backed by the European Investment Bank. While recycling could meet 20% of the EU's current permanent magnet demand of 20,000 tonnes, the EU has indicated it would legally compel companies to diversify their supply sources if voluntary efforts prove insufficient. Despite CRMA and ReSourceEU, however, many of Europe's strategic projects will struggle to meet their targets by 2030 (see figure 7), in particular recycling (12% vs 25% target), and processing (24% vs 40% target).[44]

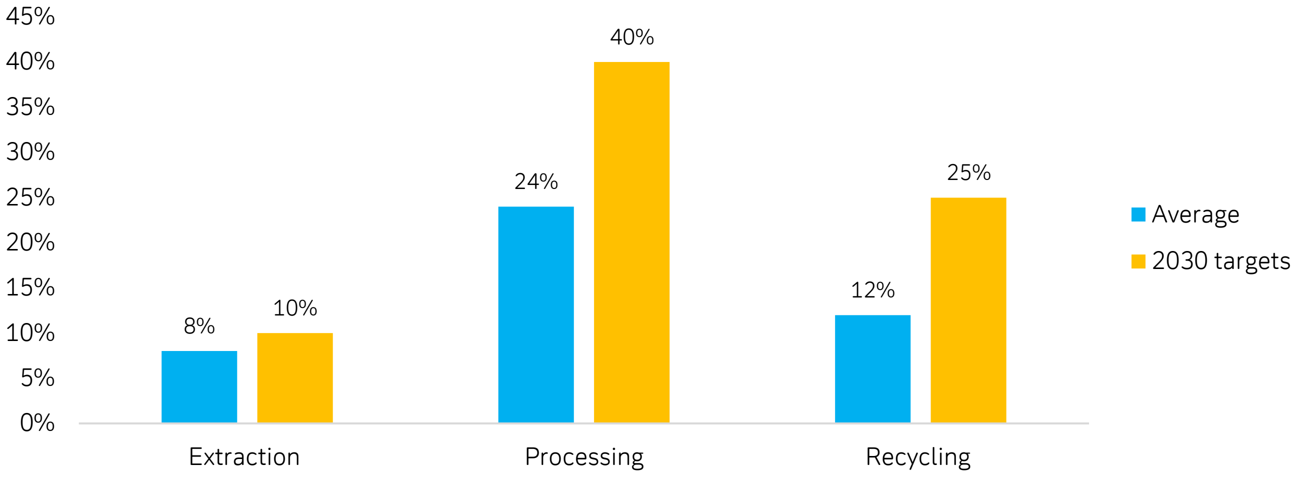
To be clear, Europe does maintain some reserves, including the Fen Complex in Norway, which remains Europe's largest known rare-earth deposit since it was discovered in 2024. And company-level initiatives are underway to restart rare-earths production, such as UK manufacturer Less Common Metals Ltd's expansion in France, Solvay in France, and LKAB in Sweden.

Europe has also undertaken several strategic partnerships with non-EU countries (e.g. Chile, Mexico, New Zealand) to boost its minerals resiliency, in a hallmark of traditional multilateralism. But as the European Court of Auditors has found, these partnerships have so far improved contribution but generated little effect on the secure supply of these raw materials.

The most pressing issue appears to be in the EU's processing abilities due to a lack of technology and a decline of facilities needed in the bloc. The ECA report notes that at present, the EU's processing capacity is shutting down, and the auditors remain uncertain whether it will actually be able to recover.[45] Our European analysts anticipate that Europe may continue to face challenges in asserting its full influence in global agenda-setting throughout 2026 as long as it remains reliant on both US and Chinese supply chains.

[40] Alice Hancock and Andy Bounds, "EU Plans Minerals Stockpile Centre to Stop US Snapping up Supplies." @FinancialTimes. Financial Times. November 19, 2025. https://www.ft.com/content/70964f99-54e9-494c-861d-139259d33893.
[41] "Special Report 04/2026: Critical Raw Materials for the Energy Transition." 2026. European Court of Auditors. 2026. https://www.eca.europa.eu/en/publications/SR-2026-04.
[42] "New Measures to Secure Raw Materials and Strengthen the EU's Economic Security," European Commission, December 3, 2025, https://commission.europa.eu/news-and-media/news/new-measures-secure-raw-materials-and-strengthen-eus-economic-security-2025-12-03_en.
[43] Ibid.
[44] "Special Report 04/2026: Critical Raw Materials for the Energy Transition." 2026. European Court of Auditors. 2026. https://www.eca.europa.eu/en/publications/SR-2026-04.
[45] "Special Report 04/2026: Critical Raw Materials for the Energy Transition." 2026. European Court of Auditors. 2026. https://www.eca.europa.eu/en/publications/SR-2026-04.

Figure 7: Average EU production capacity of specific critical minerals vs its 2030 targets



Note: Average of EU production taken from 2016 – 2020. Production capacity is the typical amount of a specific raw material that the EU can produce annually, on average, using its existing industrial facilities and resources. Source: European Court of Auditors, European Commission, Deutsche Bank Research.

3.8 Japan: A constrained success

Since facing China's export embargo in 2010, Japan has significantly reduced its reliance on Chinese rare earths from ~90% in 2010 to 60-70% today. The 2010 crisis compelled Tokyo to aggressively diversify its supply chains via long-term offtake agreements and overseas investments, notably in Australia's Lynas Rare Earths. Japan also began deep-sea rare earth mining tests in the Pacific in January 2026.[46]

However, Japan remains the world's largest rare-earth importer. In addition, new Chinese export controls in January and February 2026 — reportedly linked to Taiwan-related signalling[47] — also show that Japan is still vulnerable to disruptions from China's rare earth export controls.[48]

Japan's experience illustrates both the feasibility and the inherent limits of diversification within the framework of infrastructure realism.

[46] "Japan to Conduct Deep-Sea Rare Earth Mining Test in Pacific - Japan Today," *Japan Today*, January 8, 2026, https://japantoday.com/category/national/japan-to-conduct-deep-sea-rare-earth-mining-test-in-pacific.
[47] "China Bans Dual-Use Goods Exports for Japan Military over Taiwan Remarks," *CNBC*, January 7, 2026, https://www.cnbc.com/2026/01/07/china-bans-dual-use-goods-exports-for-japan-military-over-taiwan-remarks.html.
[48] These latest restrictions apply to a wide range of dual-use products that could contribute to Japan's military capabilities (China's definition of dual use encompasses critical chokepoints like rare earths, gallium, germanium and magnets). Unlike in 2010, China's 2026 move is explicitly intentional.

Figure 8: Select examples of the Trump Administration's critical minerals' policy

Date	Partner	Key details
April 30, 2025	Ukraine	Joint fund that will receive 50% of royalties and license fees from natural resource projects in Ukraine. The deal is more favourable to Ukraine than earlier versions negotiated between Washington and Kyiv. The US International Development Finance Corporation oversees the US side of the deal, and the US secures "first in line" preferential rights for offtake.
July 1, 2025	India, Japan, Australia	Announced the Quad Critical Minerals Initiative, which works with private sector partners to facilitate increased investments.
October 20 2025	Australia	Both governments pledged to deploy $1bn each within six months to jointly advance strategic projects. Resulting offtake will be directed to buyers in the US. Includes provisions to collaborate on supply-side interventions, including price support, streamlined permitting, geological mapping, recycling initiatives and complementary strategic stockpiling. Also includes joint mechanisms to curb the Chinese acquisition of new mining assets through tools like domestic investment screening.
October 2025	Japan, Malaysia, Thailand	The US signed MOUs with Thailand and Malaysia and a critical minerals framework with Japan. MOUs are focused on enhancing cooperation between the US and other countries over its critical minerals sectors, no funding obligation is included as of yet.
November 6, 2025	Central Asia (C5+1)	Several government-to-government and company agreements on critical minerals.
November 18, 2025	Saudi Arabia	Established a Strategic Framework for Cooperation on securing uranium, metals, permanent magnets, and critical mineral supply chains. The DoD also announced it will finance a 49% equity stake in a new rare earths refinery in SA; MP Materials will provide the technical expertise. Both countries also announced a joint civil nuclear energy cooperation. Important given Russia and China control over half of global uranium enrichment capacity, which poses serious risk to US energy security (US imports 30% of uranium from Russia).
December 4, 2025	DRC	Secures preferential access for US companies to vast cobalt, copper, and lithium reserves. The partnership was brokered alongside a peace deal between the DRC and Rwanda.
December 2025	Australia, Greece, Israel, Japan, Qatar, Korea, Singapore, UAE and UK	"Pax Silica" initiative launched by the US, which aims to build silicon and critical mineral supply chains integral to AI technologies. Focus is on private-public partnership with companies like ASML, Rio Tinto and Samsung.

Source: Deutsche Bank Research, USGS.

4. Scenarios for the critical minerals system (2026-2035)

The global critical minerals system is undergoing a structural shift. The central challenge is no longer geological availability, but rather how access, processing, and control are organized. In this new paradigm, critical minerals are moving from traded commodities toward strategic infrastructure, where power is derived from control over refining, standards, and long-term offtake rather than spot markets.

Against this backdrop, we outline four potential scenarios for 2026–2035, defined along two axes: whether supply chains converge or fragment, and whether allocation is driven primarily by state strategy or by market mechanisms (Figure 9). From 2025 onward, we find that the decades-long trend of critical minerals increasingly treated as security assets has accelerated. Over the next decade, we believe states will be more willing to use stronger policy tools—including conditional security guarantees—to secure supply and processing access. Thus, a state-driven supply chain ecosystem where minerals are used as strategic leverage a likely scenario over the next decade.

Who might emerge as a potential challenger to China? A recent CSIS comparative analysis suggests the United States is emerging as the leading global rare earths processing hub, supported by its existing infrastructure, strong R&D capacity, and bipartisan policy measures that have mobilized private investment.[49] Australia, Saudi Arabia, and Canada also show potential in this new landscape.[50]

Figure 9: Four possible scenarios for the global critical minerals system (2026 – 2035)

	Converging supply chains	Fragmenting supply chains
State-led	**Club-based resilience ●●●○**	**Minerals used as leverage ●●●●**
	- Mineral alliances (e.g. U.S.–EU–Japan–Korea–Australia) formalize coordination on stockpiles, refining capacity, and joint procurement. - China-Global South deepens ties with DRC/Indonesia/Africa/Latin America) via infrastructure and finance in exchange for resource access and processing rights. - Public investment, strategic offtakes, and critical mineral clubs emerge. - Shared ESG and tech standards reduce duplication.	- Export controls, local content mandates, and strategic embargoes. - China maintains dominance across refining; West struggles to catch up. - Producer states extract rents, tie access to security deals. - Trade becomes volatile, framed as national security. - Prices higher and more volatile; trade disputes/ WTO cases neutralised by “national security” exceptions, more financing via state-backed structures (guarantees, strategic funds, tied offtakes).
Market-led	**Open market scaling ●●○○**	**Chaotic competition ●○○○**
	- Markets respond to price signals: private investment in recycling, innovation, and substitution. - Vertical integration and long-term contracts stabilize flows. - Moderate state support (e.g., tax credits) but minimal politicization.	- Companies scramble for supply, face rising geopolitical and ESG risk. - M&A and hoarding intensify; transparency declines. - Regulatory divergence and subsidies lead to trade frictions.

Source: Deutsche Bank Research. Note: Dots indicate relative likelihood over the 2026–2035 horizon.

[49] Gracelin Baskaran and Meredith Schwartz, “Developing Rare Earth Processing Hubs: An Analytical Approach,” CSIS, 2025, https://www.csis.org/analysis/developing-rare-earth-processing-hubs-analytical-approach#h2-results.
[50] Ibid.

5. Conclusion: Critical minerals become infrastructure

This chapter has demonstrated how critical minerals have evolved from mere industrial inputs into strategic chokepoints across key technology and defense sectors. As their use-cases have evolved, countries have recognized they can no longer solely rely on market economics or corporate decision making to steer critical minerals but must designate their need as a national security priority. Furthermore, as the global system rebalances towards an "infrastructure realism" paradigm, China's control over this foundational layer creates binding bottlenecks across downstream chokepoints such as semiconductors and cloud computing. In our next chapter on Chips, we will go on to discuss the intertwined relationship between critical minerals and semiconductors, and how this has led to a dynamic of reciprocal actions and reactions, particularly between China and the US.

Still, we believe the status quo between a leading China and the rest of the world in critical minerals will unfold only gradually. The IEA projects China will continue to dominate mineral refining (76%) and mining (51%) through 2030.[51] Japan required more than a decade to reduce its direct reliance on Chinese rare earths from 90% to 60% — illustrating the time, cost, and persistence required to unwind dependence once industrial ecosystems are entrenched.

Managed interdependence, therefore, will likely define the rare earths and critical minerals landscape for the next decade, as the US and other countries attempt to replicate the industrial edge Beijing began building in the 1990s. Returning to the "Mineral Fordism" analogy, historian Stefan Link observed that "all late developers must turn, for capital and technology, to those they seek to emulate" —a path that is structurally more difficult in today's geopolitically fragmented system.

As this slow transformation unfolds, the use of minerals for geopolitical leverage is likely to become an established feature of the global system, with China selectively deploying its mineral sector capabilities to advance trade, security, and foreign policy goals. In an infrastructure-realist world, control over processing, standards, and access—not ownership alone—will increasingly define power. In light of this, over the next five years, we expect geopolitical friction related to critical minerals to manifest in several areas: (i) ongoing competition between China and the US over critical minerals; and (ii) competition between China and the rest of the world over critical minerals; and (iii) competition between the US and the rest of the world, as it continues to pursue its dual goals of gaining access to strategically important regions and ensuring critical mineral security.

[51] "Rare earth elements – Analysis", IEA, May 21, 2025, https://www.iea.org/reports/rare-earth-elements-2.

Appendix 1

This material has been prepared by the Deutsche Bank Research Institute and is provided to you for general information purposes only. The Institute leverages the views, opinions, and research from Deutsche Bank Research, economists, strategists, and research analysts. Accordingly, you should assume that content in this document is based on or was previously published and provided to Deutsche Bank clients who may have already traded on the basis of it.

Any views or estimates expressed in this material reflect the current views of the author(s) and may differ from the views and estimates of Deutsche Bank AG, its affiliates, other Deutsche Bank personnel, and other materials published by Deutsche Bank. The content in this material is valid as of the date shown on the first page and may change without notice. Deutsche Bank has no obligation to provide any updates or changes to the information herein.

This material should not be used as a basis for trading securities or other financial products and should not be considered to be a recommendation or individual investment advice for any particular person. It does not constitute an offer, solicitation, or an invitation by or on behalf of Deutsche Bank to any person to buy or sell any security or financial instrument. Nothing in this material constitutes investment, legal, accounting or tax advice. Deutsche Bank engages in securities transactions, including on a proprietary basis, and may do so in a manner inconsistent with the views or information expressed in this material.

While information in this material has been obtained from sources believed to be reliable, neither Deutsche Bank AG nor any of its affiliates makes any representation or warranty, express or implied, as to the accuracy or completeness of the statements or any information contained in this material and therefore any liability is expressly disclaimed. This material is provided without any obligation, whether contractual or otherwise. Information regarding past transactions or performance is not indicative of future results.

In the U.S. this report is approved and/or distributed by Deutsche Bank Securities Inc., a member of FINRA. In Germany this information is approved and/or communicated by Deutsche Bank AG Frankfurt, licensed to carry on banking business and to provide financial services under the supervision of the European Central Bank (ECB) and the German Federal Financial Supervisory Authority (BaFin). In the United Kingdom this information is approved and/or communicated by Deutsche Bank AG, London Branch, a member of the London Stock Exchange, authorized by UK's Prudential Regulation Authority (PRA) and subject to limited regulation by the UK's Financial Conduct Authority (FCA) (under number 150018) and by the PRA. This information is distributed in Hong Kong by Deutsche Bank AG, Hong Kong Branch, in Korea by Deutsche Securities Korea Co. and in Singapore by Deutsche Bank AG, Singapore Branch. In Japan this information is approved and/or distributed by Deutsche Securities Limited, Tokyo Branch. In Australia, retail clients should obtain a copy of a Product Disclosure Statement (PDS) relating to any financial product referred to in this report and consider the PDS before making any decision about whether to acquire the product.

By accessing this material, you agree that its content may not be reproduced, distributed or published by any person for any purpose, in whole or part, without Deutsche Bank's prior written consent. You also agree that you shall not scrape, extract, download, upload, sell or offer for sale any of the content in this material, and you agree not to use, or cause to be used, any computerized or other manual or automated program or mechanism, tool, or process, including any scraper or spider robot, to access, extract, download, scrape, data mine, display, transmit, or publish, any of the content in this material.